

10.14 part 1.

$$\underline{A} = \left\{ \frac{1}{2} + \frac{0.3}{4} + \frac{0.5}{6} + \frac{0.2}{8} \right\}$$

$$\underline{B} = \left\{ \frac{0.5}{2} + \frac{0.4}{4} + \frac{0.1}{6} + \frac{1}{8} \right\}$$

$$\underline{A} \cup \underline{B} = \max\{\mu_A, \mu_B\} = \left\{ \frac{1}{2} + \frac{0.4}{4} + \frac{0.5}{6} + \frac{1}{8} \right\}$$

$$\underline{A} \cap \underline{B} = \min\{\mu_A, \mu_B\} = \left\{ \frac{0.5}{2} + \frac{0.3}{4} + \frac{0.1}{6} + \frac{0.2}{8} \right\}$$

$$\overline{\underline{A}} = 1 - \mu_A(x) = \left\{ \frac{0}{2} + \frac{0.7}{4} + \frac{0.5}{6} + \frac{0.8}{8} \right\}$$

eg: Fuzzy $\underline{A}_1 = \left\{ \frac{1}{2} + \frac{0.3}{4} + \frac{0.5}{6} + \frac{0.2}{8} + \frac{0}{9} \right\}$

different

$$\underline{A|B} = \underline{A} \cap \overline{\underline{B}} = \left\{ \frac{0.5}{2} + \frac{0.3}{4} + \frac{0.5}{6} + \frac{0}{8} \right\}$$

10.15

$$\underline{A} = \left\{ \frac{0.2}{1} + \frac{0.3}{2} + \frac{0.4}{3} + \frac{0.5}{4} \right\}$$

$$\underline{B} = \left\{ \frac{0.1}{1} + \frac{0.2}{2} + \frac{0.2}{3} + \frac{1}{4} \right\}$$

$$\mu_{A+B}(x) = [\mu_A(x) + \mu_B(x)] - [\mu_A(x) \cdot \mu_B(x)]$$

$\rightarrow \frac{1}{4}$ [0.1]
1.5 - 1 = 0.5

$$= \left\{ \frac{0.1}{1} + \frac{0.5}{2} + \frac{0.6}{3} + \frac{0.5}{4} \right\} - \left\{ \frac{0.02}{1} + \frac{0.06}{2} + \frac{0.08}{3} + \frac{0.3}{4} \right\}$$

$$= \left\{ \frac{0.44}{2} + \frac{0.52}{3} + \frac{0}{4} \right\}$$

$$\mu_{A \cdot B}(x) = \mu_A(x) \cdot \mu_B(x) \text{ 如上}$$

Bounded sum 有限和

$$\mu_{A \oplus B}(x) = \min[1, \mu_A(x) + \mu_B(x)]$$

$$= \min \left[1, \left\{ \frac{0.3}{1} + \frac{0.5}{2} + \frac{0.6}{3} + \frac{0.5}{4} \right\} \right]$$

$$= \left\{ \frac{0.3}{1} + \frac{0.5}{2} + \frac{0.6}{3} + \frac{0.5}{4} \right\}$$

Bounded difference

$$\mu_{A \ominus B}(x) = \max[0, \mu_A(x) - \mu_B(x)]$$

$$= \left\{ \frac{0.1}{1} + \frac{0.1}{2} + \frac{0.2}{3} + \frac{0.5}{4} \right\}$$

$$\rightarrow |0.5 - 1|$$

part 2. 经典关系与模糊关系

笛卡尔乘积 Cartesian product

$$X \times Y = \{ (x, y) \mid x \in X \text{ and } y \in Y \} \quad \text{if } (x \neq y) \text{ then } x \times y \neq y \times x$$

Fuzzy Relation

$$\underline{R} = \{ (x, y), \underline{\mu_R(x, y)} \mid (x, y) \in X \times Y \}$$

$$\underline{A} = \left\{ \frac{0.3}{x_1} + \frac{0.7}{x_2} + \frac{1}{x_3} \right\} \quad \underline{R} = \underline{A} \times \underline{B}$$

$$\underline{B} = \left\{ \frac{0.4}{y_1} + \frac{0.9}{y_2} \right\}$$

$$R = \begin{matrix} & y_1 & y_2 \\ \begin{matrix} x_1 \\ x_2 \\ x_3 \end{matrix} & \begin{bmatrix} 0.3 & 0.3 \\ 0.4 & 0.7 \\ 0.4 & 0.9 \end{bmatrix} \end{matrix}$$

$$\mu_R(x_i, y_j) = \min[\mu_A(x_i), \mu_B(y_j)] \\ = 0.3$$

0.20

$X \times Y$

$$R(x, y) = \begin{cases} 0 & (x, y) \notin R \\ 1 & (x, y) \in R \end{cases}$$

$$R \subset T \Rightarrow \mu_R(x, y) \leq \mu_T(x, y)$$

复合函数 $\max_{x \in X} \min_{y \in Y} (f(x, y))$ 数据库的数字基础

$f(x)$

$$\max\text{-}\min = \{ (x_1, y_1), (x_2, y_2), (x_3, y_3) \} = \min \{ \dots \} \\ = \max$$

$$T = R \circ S$$

$$X_T(x, z) = \bigvee_{y \in Y} (X_R(x, y) \wedge X_S(y, z)) = \bigvee_{y \in Y} (X_R(x, y) \cdot X_S(y, z))$$

Fuzzy Max-min

Max Product

$$\underline{R} = \begin{matrix} & y_1 & y_2 \\ \begin{matrix} x_1 \\ x_2 \end{matrix} & \begin{bmatrix} 0.6 & 0.3 \\ 0.2 & 0.9 \end{bmatrix} \end{matrix}$$

$$\underline{S} = \begin{matrix} & z_1 & z_2 & z_3 \\ \begin{matrix} y_1 \\ y_2 \end{matrix} & \begin{bmatrix} 1 & 0.5 & 0.3 \\ 0.8 & 0.4 & 0.7 \end{bmatrix} \end{matrix}$$

$$T = R \circ S = \begin{matrix} & z_1 & z_2 & z_3 \\ x_1 & \begin{bmatrix} 0.6 & 0.5 & 0.3 \end{bmatrix} \\ x_2 & \begin{bmatrix} 0.8 & 0.4 & 0.7 \end{bmatrix} \end{matrix}$$

$$\mu_T(x_i, z_j) = \max \{ \min[\mu_R(x_i, y_1), \mu_S(y_1, z_j)], \min[\mu_R(x_i, y_2), \mu_S(y_2, z_j)] \}$$

$$\underset{\sim}{T} = \underset{\sim}{R} \circ \underset{\sim}{S} = \begin{matrix} & \underset{\sim}{z}_1 & \underset{\sim}{z}_2 & \underset{\sim}{z}_3 \\ \underset{\sim}{x}_1 & \begin{bmatrix} 0.60 & 0.50 & 0.20 \end{bmatrix} \\ \underset{\sim}{x}_2 & \begin{bmatrix} 0.72 & 0.36 & 0.63 \end{bmatrix} \end{matrix}$$

$$\mu_{\sim T}(x_i, z_j) = \max \{ [\mu_{\sim R}(x_i, y_1) \cdot \mu_{\sim S}(y_1, z_j)], [\mu_{\sim R}(x_i, y_2) \cdot \mu_{\sim S}(y_2, z_j)] \}$$

离散数学 

$$(x_i, x_i) \in R. \text{ or } X_R(x_i, x_i) = 1 \quad \text{自反.}$$

$$X_R(x_i, x_j) = X_R(x_j, x_i) \quad \text{对称对称}$$

reflexive. symmetry. Transitivity.
反身性 对称性 传递性

$$\mu(x_i, x_k) = \lambda_1, \quad \mu(x_k, x_j) = \lambda_2. \quad \mu(x_i, x_j) = \lambda$$

$$\lambda \geq \min(\lambda_1, \lambda_2) \quad \text{则具有传递性.}$$

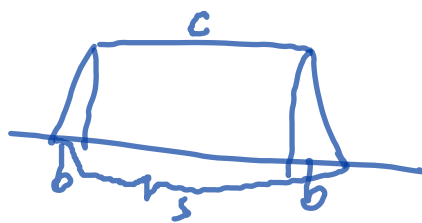
10.21

Membership Function

(1) Core $\mu(x) = 1$

(2) Support $\mu(x) > 0$

(3) Boundary $(0 < \mu(x) < 1)$



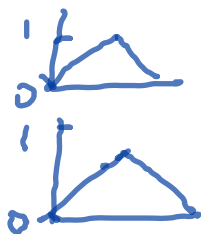
fuzzy set.

(1) Normal

(2) subnormal

(3) convex (对称性)

(4) Non convex



$[0, 1]$

λ -cut.

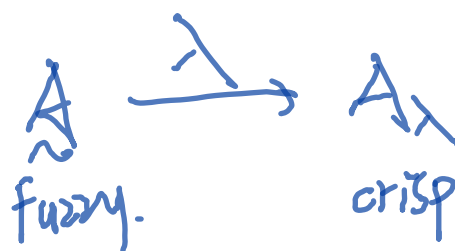
$$A_\lambda = \{x \mid \mu_A(x) \geq \lambda\}$$

① $U = U$

② $A = A$

③ $(\bar{A})_\lambda \neq (\bar{A}_\lambda)$ 降 $\lambda = 0.5$

$$\lambda\text{-cut. } A_\lambda = \begin{cases} 1 & \geq \lambda \\ 0 & < \lambda \end{cases}$$



类似 k-means 聚类后

聚类.

$$(\bar{A})_\lambda = \{x \mid \mu_{\bar{A}}(x) \geq \lambda\} = \left\{ \frac{0.8}{x_1} + \frac{0.7}{x_2} + \frac{0.6}{x_3} + \frac{0.3}{x_4} + \frac{0.1}{x_5} \right\}$$

$$(\bar{A})_{0.7} = \{x_1, x_2, x_3\}$$

$$A_\lambda = \{x \mid \mu_A(x) \geq \lambda\}$$

先非再并

